

Note 6.3 Diagonalization

Intro

Theorem 6.3.1

For k distinct eigenvalues of an $n \times n$ matrix A with corresponding k eigenvectors, then this eigenvectors are linearly independent.

It can be proved by the definition of linearly independent and contradicts.

Definition

An $n \times n$ matrix A is diagonalizable if there exists a nonsingular matrix X and a diagonal matrix D such that

$$X^{-1}AX = D$$

and we say X diagonalizes A .

Theorem 6.3.2

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

And the column vectors of X is n eigenvectors of A and the elements of D is the n corresponding eigenvalues.

Remarks

- Diagonalizing matrix is not unique since it can be modified by reordering the columns or multiplying them by nonzero scalars.
- If A has n distinct eigenvalues, it must be diagonalizable. If not, it are not sure, depending on whether A has n distinct eigenvectors.
- $A^k = XD^k = X^{-1}$, where the elements in D^k is λ_i^k .
- If A has fewer than n linearly independent eigenvectors, it is defective.

Sometimes one eigenvalues corresponds to more eigenvectors, which depends on the nullity of $A - \lambda I$.

The Exponential of a Matrix

For a matrix exponential e^A , it can be expressed by the convergent power series

$$e^A = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \dots$$

For a diagonal matrix D with λ_i as elements, then e^D has the elements e^{λ_k} .

Thus if A is diagonalizable, then $e^A = Xe^DX^{-1}$.

And it can be used to solve initial value problem

$$Y' = AY, Y(0) = Y_0$$

and its solution is

$$Y = e^{tA}Y_0$$